Integration and Evolvability of Simian Cranial Sexual Dimorphism

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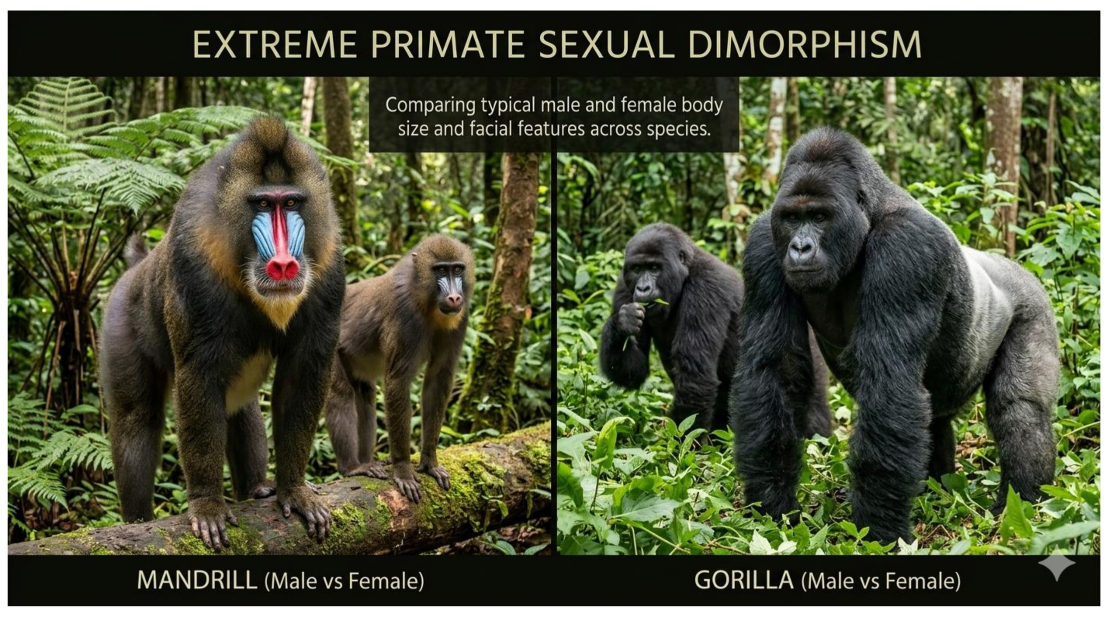
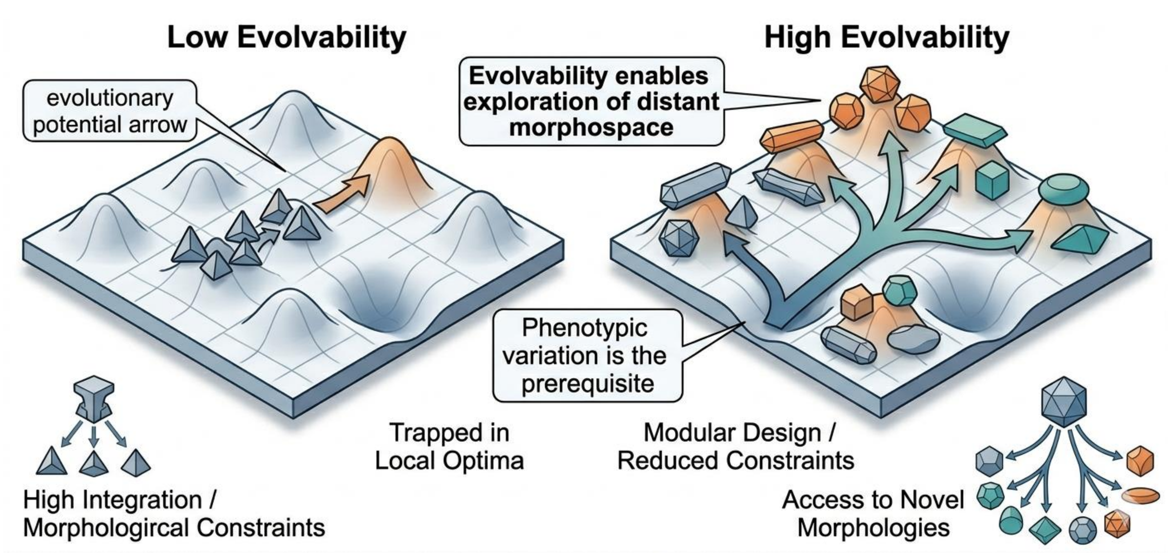
Sexual Dimorphism



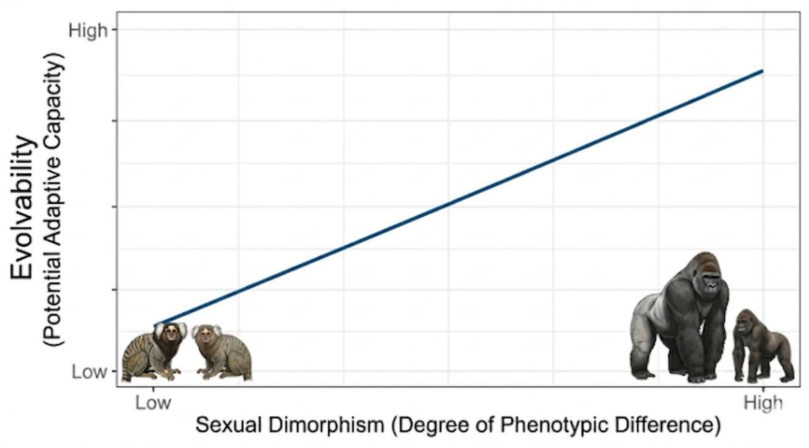
- Sexual dimorphism shows extensive interspecific variation
- The evolution of sexual dimorphism is likely multifactorial, but the role of evolvability remains unclear

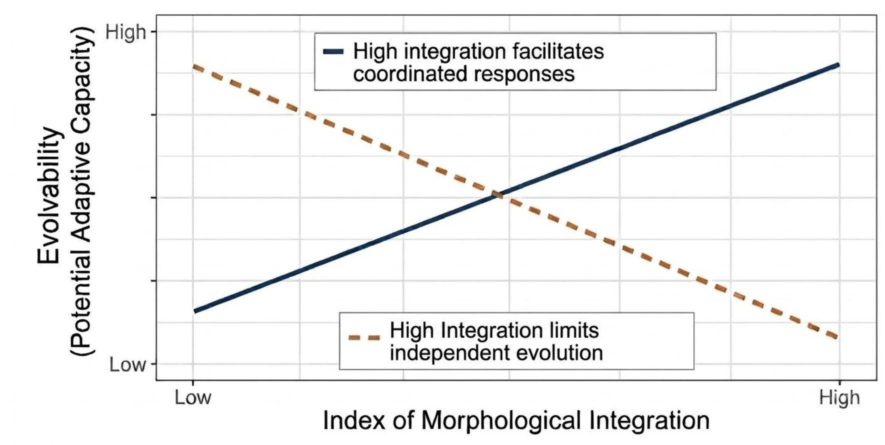
Evolvability and Modularity

- Low evolvability may slow the pace of evolutionary change
- High evolvability may promote greater morphological diversification and faster evolutionary change



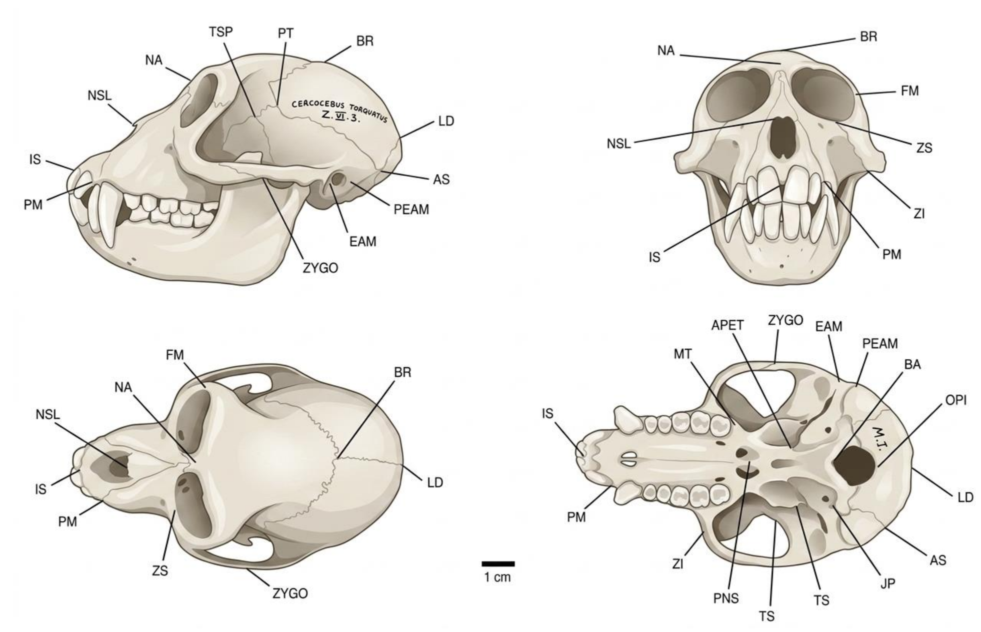
Hypothesis

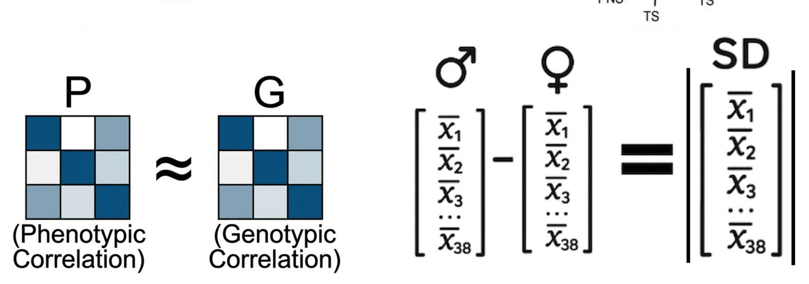




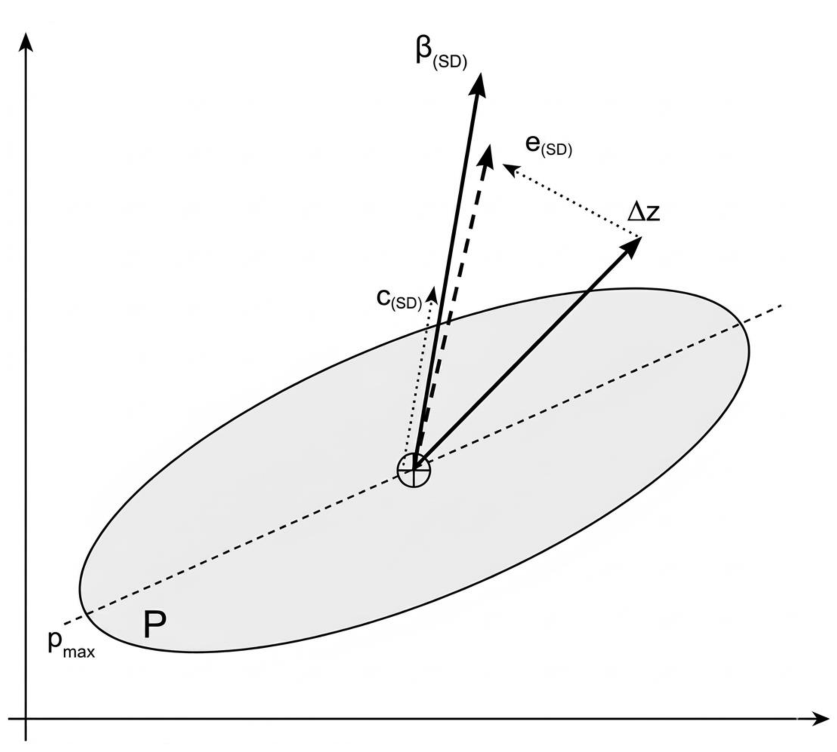
Data

- 50 species from two simian clades: Platyrrhini and Catarrhini
- 38 linear cranial measurements collected from simian skulls
- ~ 5.416 simian skulls measured
- **P** matrices estimated for each species
- Sexual dimorphism was measured as the **norm** of the vector difference between males and females





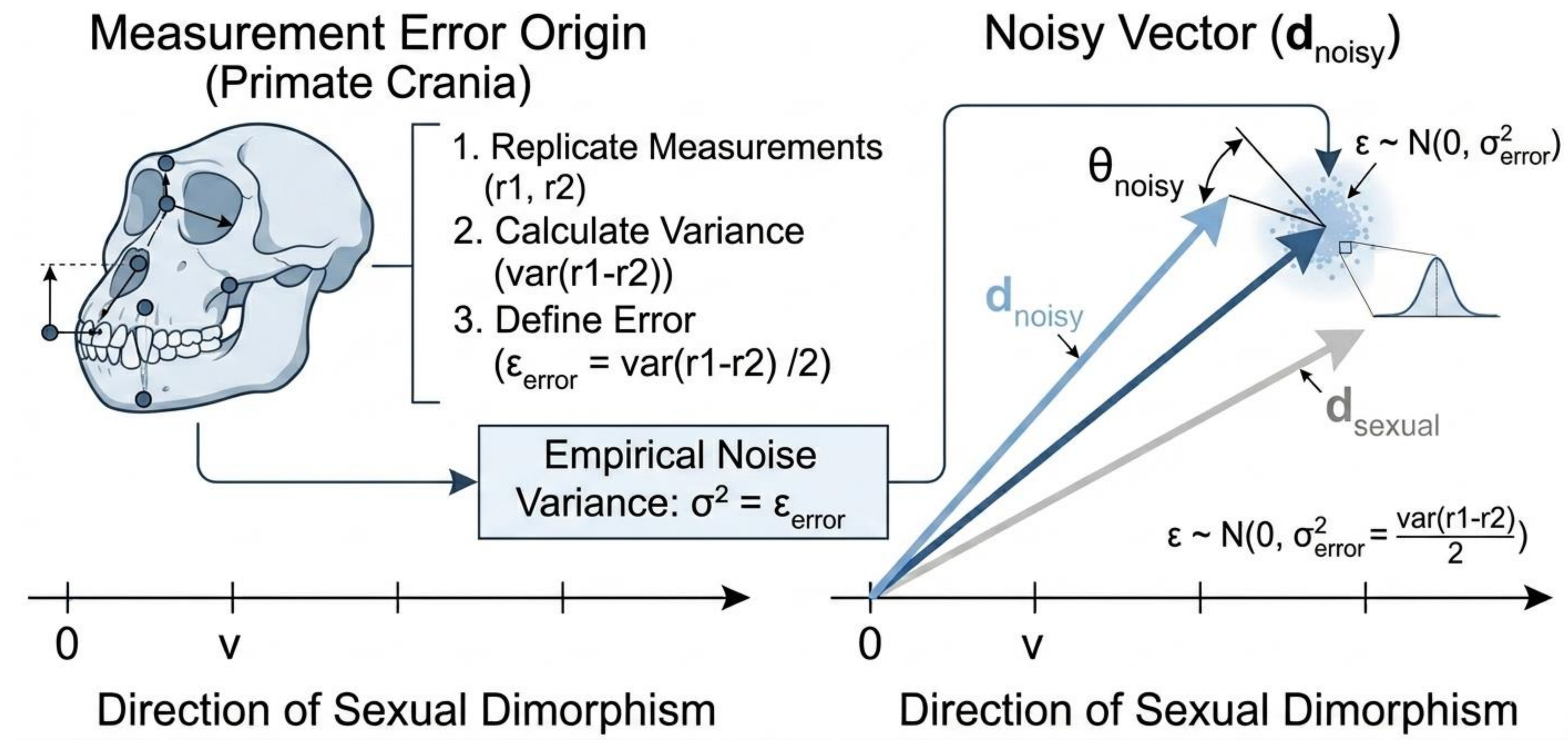
Analysis



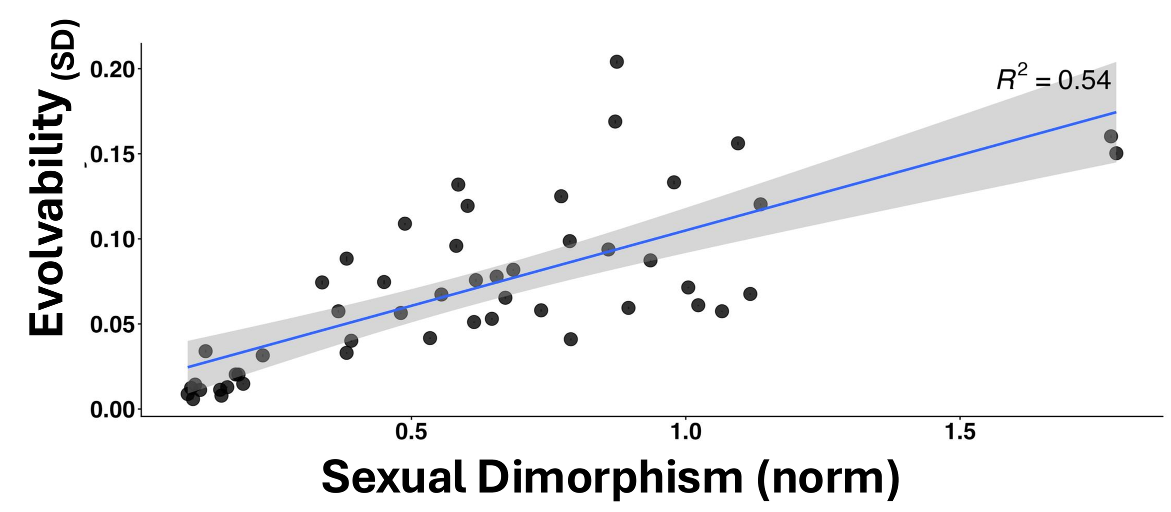
$$\text{evolvability}_{(SD)} = \frac{(\mathrm{SD} + \varepsilon)'\mathbf{P}\,\mathrm{SD}}{|\mathrm{SD} + \varepsilon|^2}$$

$$\text{conditional evolvability}_{(SD)} = \frac{1}{\mathrm{SD}^{\mathrm{T}}\mathbf{P}^{-1}\mathrm{SD}}$$

$$\text{constraint}_{(SD)} = 1 - \frac{\text{conditional evolvability}_{(SD)}}{\text{evolvability}_{(SD)}}$$

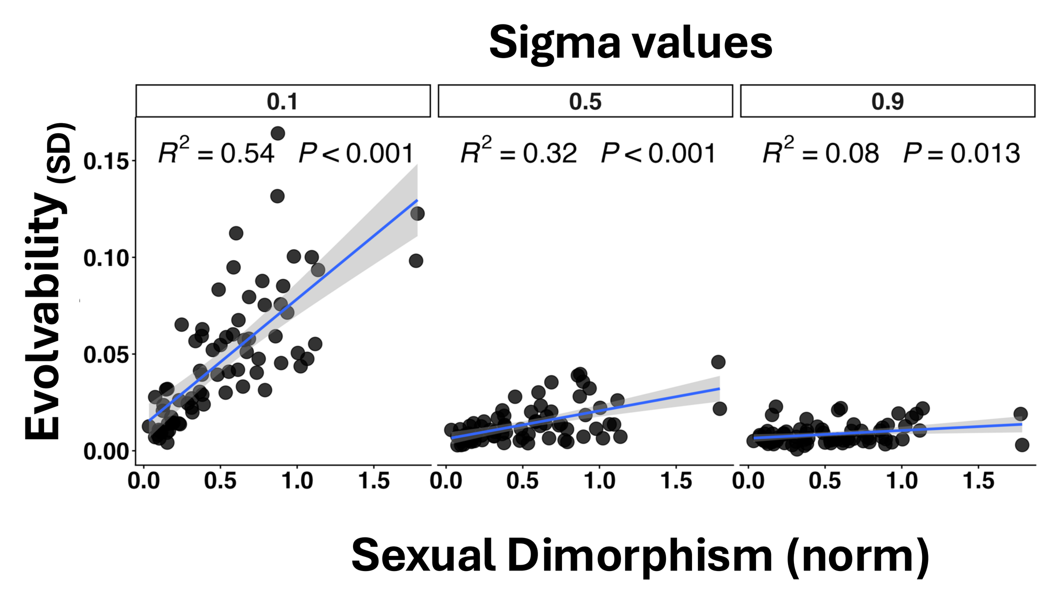


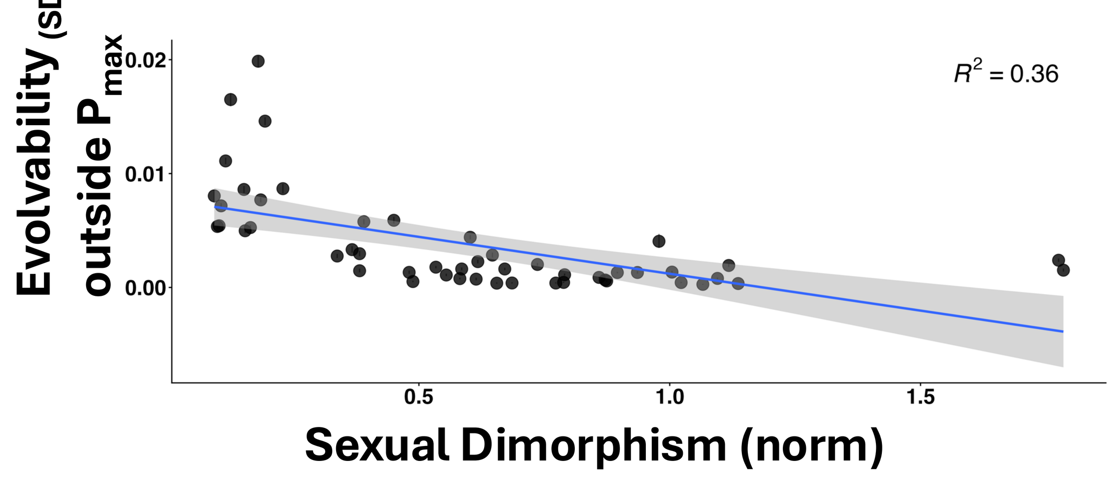
Results



- Greater sexual dimorphism is associated with higher evolvability

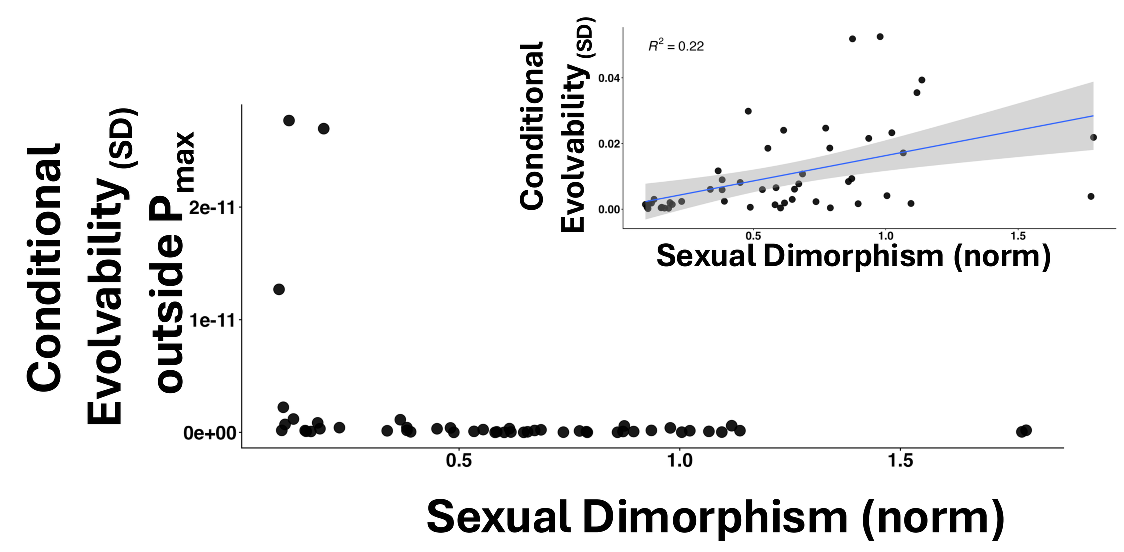
- High measurement error weakens and may eliminate the relationship between sexual dimorphism and evolvability

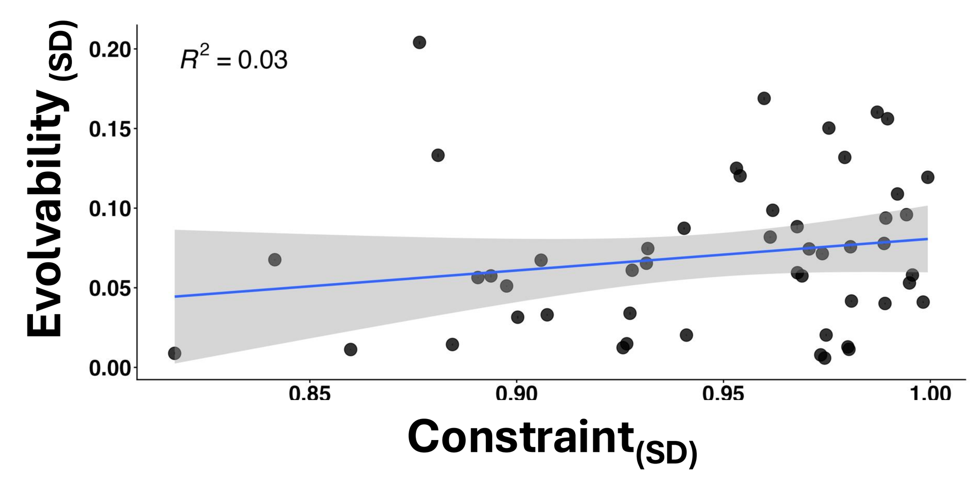




- Excluding $\mathbf{P}_{max}$ changes the relationship, making it nonlinear and inverse

- Conditional evolvability of sexual dimorphism drops to practically zero after controlling for $\mathbf{P}_{max}$





- No relationship was found between evolvability and constraint
- Removing $\mathbf{P}_{max}$ results in constraint values close to the theoretical maximum (1)

Conclusions and Next Steps

- Sexual dimorphism tends to evolve along highly evolvable directions
- Constraint alone does not influence evolvability among species
- Measurement error is important but does not substantially alter the observed patterns

- Test whether these patterns are phylogenetically widespread
- Better examine the consequences of removing **Pmax** on evolvability estimates.

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